

# INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

Department of Computer Science and Engineering



## M.Tech Project Report

(MTP Phase II)

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### Anonymity Preserving Decoupled Distributed Authentication for Multihop IoT Networks

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# CERTIFICATE

*This is to certify that the work contained in this thesis entitled “**Anonymity Preserving Decoupled Distributed Authentication for Multihop IoT Networks**” is a bonafide work of **Anup Kulkarni** (Roll No. 244101005), carried out in the Department of Computer Science and Engineering, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.*

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# Abstract

The primary objective of this work is to identify, address, and validate critical security and privacy gaps in DAuth, a lightweight decoupled distributed authentication scheme for multihop IoT networks. Two significant vulnerabilities were identified. First, the real device identity is transmitted in plaintext over the public channel during every authentication session. This exposure enables any passive adversary to track devices across sessions, correlate repeated authentication attempts, and infer network behaviour, which directly violates modern privacy requirements. Second, when a key-exchange packet is lost in transit, the device and authentication server hold divergent session states, causing all subsequent authentication attempts to fail and forcing the device through a costly full re-enrolment.

To address both vulnerabilities, this work proposes an extension of DAuth that preserves device anonymity and tolerates desynchronisation. Per-session pseudonym rotation replaces the real device identity in all open-channel communications with a rotating pseudonym that is refreshed after each successful key exchange, so that successive pseudonyms are computationally unlinkable. A dual-state mechanism maintained at the authentication server retains both the current and the immediately preceding session state, enabling seamless recovery from a single packet-loss event without re-enrolment. Formal verification under a standard adversary model confirms authentication correspondence, session key secrecy, forward secrecy, anonymity preservation, and offline guessing resistance. COOJA/Contiki-NG simulations on a 100-node RPL network show a 9.5 % reduction in per-device energy over LAAKA and 32.4 % over Zhou et al., with consistent gains across all network sizes and authentication server pool configurations. Hardware evaluation on Raspberry Pi 4B confirms that the proposed scheme attains virtually identical cost to the non-anonymous DAuth baseline (within 1 %), while reducing energy by approximately 36 % and 43 % over LAAKA and Zhou et al., respectively, demonstrating that provable anonymity is achieved at no measurable overhead on real IoT devices.





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# Chapter 1

## Introduction

The Internet of Things (IoT) has expanded rapidly and now underpins critical applications across smart manufacturing, power grids, remote health monitoring, and intelligent transport systems. These deployments rely on large numbers of resource-constrained devices communicating over shared wireless channels that are inherently exposed to eavesdropping. Securing such networks against impersonation, replay, man-in-the-middle, key compromise, and device-capture attacks is therefore a fundamental operational requirement.

Considerable research has addressed IoT device authentication through mechanisms tailored to the severe power and memory constraints of embedded hardware. Physical Unclonable Functions (PUFs), hash-based challenge-response protocols, and symmetric-key constructions have emerged as dominant lightweight approaches owing to their low computational overhead. DAuth [1] follows this direction: it combines PUFs with a hash-based accumulator, eliminates long-term key storage, distributes authentication across multiple in-network nodes, and supports multihop verification at minimal overhead.

Despite these properties, a fundamental privacy gap persists. During each authentication session, the device’s real identity  $ID_D$  is transmitted in plaintext over the public channel. A passive adversary can capture these transmissions to track the device across sessions, correlate repeated authentication attempts, and infer network behaviour, which

directly violates modern privacy requirements.

Identity privacy has been established as a fundamental property in IoT authentication, not merely an optional feature. Survey literature confirms that anonymity, untraceability, and forward secrecy are essential, particularly when devices operate in open or adversarial environments [2]. Existing countermeasures such as session-refreshed pseudonyms and hash-chain-based temporary identities partially address this exposure but often introduce new challenges such as desynchronisation, excessive storage overhead, slow identity lookup, or increased computation. The sub-framework taxonomy of [2], covering SF1-UA through SF9-FS, illustrates how difficult it is to simultaneously satisfy anonymity, unlinkability, and resilience to message loss.

This work investigates how anonymity can be integrated into DAuth without compromising its lightweight properties. All authentication messages of DAuth traverse an open channel, and transmitting  $ID_D$  in plaintext exposes the device to tracking, session linkage, and targeted impersonation. Drawing on prior anonymous-authentication literature, this work identifies a low-overhead pseudonym mechanism that integrates cleanly into DAuth. The objective is identity anonymity, session unlinkability, and non-traceability, achieved without relaxing the lightweight and distributed properties of DAuth.

A second gap in DAuth concerns desynchronisation resilience. Because DAuth retains only a single session nonce at the authentication server, any loss of the key-exchange response packet causes the device and server states to diverge, leading to a permanent authentication failure that can only be resolved through costly full re-enrolment.

## 1.1 Contributions

This work makes the following key contributions.

1. An anonymity-preserving scheme is proposed that introduces per-session pseudonym rotation to preserve device anonymity and a dual-state mechanism to recover from

desynchronisation without re-enrolment.

2. The security of the proposed scheme is validated through both informal analysis and formal verification using ProVerif.
3. The efficiency of the proposed scheme is evaluated using theoretical analysis, COOJA/Contiki-NG simulation, and hardware experiments on Raspberry Pi 4B, covering communication overhead, energy consumption, and CPU utilisation.

## **1.2 Organisation of The Report**

The rest of this report is organised as follows. Chapter 2 surveys the related literature. Chapter 3 describes the prerequisite concepts. Chapter 4 overviews DAuth [1]. Chapter 5 presents the proposed scheme in full detail. Chapter 6 contains the security analysis. Chapter 7 presents the performance evaluation. Chapter 8 concludes the report.



# Chapter 2

## Literature Survey

IoT authentication research has increasingly favoured lightweight and distributed designs, driven by the prohibitive load that centralised systems impose on a single gateway and the single point of failure they introduce. Given the severe power and memory constraints of IoT devices, the research community has sought architectures in which multiple network nodes share the authentication workload.

**Light-SAE** [3] exemplifies this trend. It enables already-authenticated nodes to assist newcomers through a simple challenge-and-response join protocol, using TinyJAMBU for lightweight encryption and a modified Diffie–Hellman exchange to establish the session key in multihop networks, thereby reducing the load on the gateway. However, Light-SAE transmits node identities in plaintext and provides no protection for anonymity or session unlinkability, rendering it unsuitable for privacy-sensitive deployments.

**SAKMS** [4] targets industrial IoT systems operating under the 6TiSCH schedule, employing ECC-based mutual authentication with ECQV certificates, formalised key management, and strong forward secrecy, with correctness confirmed by ProVerif. Its reliance on multiple elliptic-curve operations, however, imposes overhead that exceeds what severely constrained IoT nodes can sustain. SAKMS also embeds fixed device identifiers within messages, leaving devices susceptible to tracking across sessions. These limitations illus-

trate that cryptographic strength alone does not imply privacy preservation or lightweight execution.

The **DAuth PUF-based authentication scheme** [1] addresses many of these weaknesses. It eliminates long-term secret storage by deriving device secrets from PUFs, uses a hash-based accumulator for membership verification, and decouples the authentication server from the parent node so that both parent and non-parent nodes can serve as authenticators, distributing load in multihop networks. Its computational primitives (hash, XOR, and two AES calls) keep the per-device footprint minimal. Its primary limitation is that the device identity ( $ID_D$ ) is transmitted directly in every authentication exchange; since all messages traverse a public channel, a passive observer can capture the identity, track the device over time, and mount targeted attacks. Identity concealment is therefore a necessary extension to this protocol.

Several anonymous authentication schemes have been proposed to address such privacy concerns. **He et al.** [5] achieve lightweight anonymous authentication using hash-and-XOR primitives for CoAP/OSCORE but rely on a centralised server, forfeiting the decoupled-distribution advantage. **LAACA** [6] protects device identity during mutual authentication using only lightweight hash and XOR operations; its overhead, however, stems from larger message payloads and additional message exchanges on severely constrained nodes.

A comprehensive survey on lightweight anonymous authentication [2] catalogues approaches including temporary pseudonyms, PUF-assisted anonymous login, hidden membership proofs via accumulators, and unlinkable tokens, while documenting common pitfalls such as desynchronisation, pseudonym exhaustion, and inadequate replay protection. Their taxonomy deconstructs existing protocols into nine sub-frameworks (SF1-UA through SF9-FS) and two general frameworks (GF1, GF2), finding that only GF2 simultaneously achieves user anonymity, forward secrecy, and desynchronisation resistance.

Building on this taxonomy, the present work adopts **GF2** as its target architectural pattern and realises a *modified* instantiation of it tailored to resource-constrained multihop

IoT nodes. Concretely, the proposed scheme combines the user-anonymity sub-frameworks (SF3-UA, SF5-UA, SF6-UA), which use rotating pseudonyms instead of a static identity, with the forward-secrecy sub-frameworks (SF7-FS, SF8-FS, SF9-FS), which derive each session key from independently refreshed material. Unlike the survey’s reference constructions that rely on heavy cryptographic primitives, the proposed scheme uses lightweight PUF-based secret derivation and hash-XOR masking, and meets GF2’s desynchronisation-resistance requirement through a dual-state recovery mechanism at the authentication server. The result is a GF2-compliant design with the same low overhead as DAuth [1].

**Zhou et al.** [7] propose a security-enhanced lightweight and anonymity-preserving user authentication scheme for IoT-based healthcare, providing hash-based anonymous authentication with nonce masking to protect sensitive metadata over open wireless channels.

Other works use **blockchain** [8] and **certificateless cryptography** [9] to achieve decentralisation and identity hiding, but they incur prohibitive storage and consensus overhead. **Li et al.** [10] and **Zheng et al.** [11] combine PUF with session key exchange but rely on ECC operations too expensive for highly constrained nodes.

Despite this body of work, no existing scheme simultaneously satisfies all of the following requirements: (i) decoupled authentication, (ii) PUF-based secrecy without long-term key storage, (iii) low-overhead hash-XOR computation, and (iv) device anonymity with session unlinkability and desynchronisation recovery. This gap motivates the present work, which builds an anonymous and desynchronisation-resilient extension of DAuth [1] while preserving its lightweight and distributed properties.

To show the differences between all these methods clearly, Table 2.1 lists the use of decoupling, PUF, accumulator-based authentication, authentication secret update, anonymity, desynchronisation recovery, and the computation and communication overhead for all the schemes studied.

**Table 2.1:** Features Comparison of Authentication Schemes.

| Scheme            | Dec. | PUF | Acc. | ASU | Anon. | Desync | C.OH     | M.OH     |
|-------------------|------|-----|------|-----|-------|--------|----------|----------|
| RAAF-MEC [12]     | ×    | ✓   | ×    | ✓   | ×     | ×      | H        | L        |
| SAKMS [4]         | ✓    | ×   | ×    | ×   | ×     | ×      | H        | L        |
| Li et al. [10]    | ✓    | ✓   | ×    | ✓   | ×     | ×      | H        | L        |
| Zheng et al. [11] | ✓    | ✓   | ×    | ✓   | ×     | ×      | M        | L        |
| LAACA [6]         | ×    | ×   | ×    | ✓   | ✓     | ×      | L        | L        |
| Zhou et al. [7]   | ×    | ✓   | ×    | ✓   | ✓     | ×      | H        | M        |
| DAuth [1]         | ✓    | ✓   | ✓    | ✓   | ×     | ×      | L        | L        |
| <b>Proposed</b>   | ✓    | ✓   | ✓    | ✓   | ✓     | ✓      | <b>L</b> | <b>L</b> |

*Dec.*: Decoupled design. *Acc.*: Hash-based accumulator. *ASU*: Auth. secret update.

*Anon.*: Device anonymity via pseudonym. *Desync*: Desync. recovery.

*C.OH*: Computation: L (Hash/XOR/ $\leq 2$  AES), M ( $> 2$  AES), H (ECC/Pairings).

*M.OH*: Message exchanges: L (1–3), M (4–5), H ( $> 5$ ).



# Chapter 3

## Prerequisites for Objective

### 3.1 Physically Unclonable Function

A Physically Unclonable Function (PUF) is a one-way physical function that generates a unique response  $R$  for a given challenge  $C$  from the challenge space  $\mathcal{C}$ . PUFs exploit manufacturing variations in integrated circuits that are practically impossible to replicate. Various types of PUFs have been proposed, including SRAM PUFs, Arbiter PUFs, and Ring Oscillator (RO) PUFs. In this work, a Ring Oscillator PUF is used due to its higher execution speed and lower overhead.

A function is considered a valid and secure PUF if it satisfies the following properties:

- **Evaluatable:** A legitimate entity can easily compute a response  $R \in \mathcal{R}$  for any given challenge  $C \in \mathcal{C}$ .
- **Unpredictability:** An adversary cannot predict the response  $R$  for any challenge  $C$  without physical access to the PUF.
- **Uniqueness:** For two different PUF instances,  $\text{PUF}_1$  and  $\text{PUF}_2$ , the probability that  $\text{PUF}_1(C) = \text{PUF}_2(C)$  for the same challenge  $C$  is negligible.
- **Reliability:** For a fixed challenge  $C$ , the PUF response  $\text{PUF}(C) = R$  remains stable

over time and environmental variations.

- **One-wayness:** Given a response  $R$ , it is computationally infeasible to determine the corresponding challenge  $C$  or invert the PUF.

These properties make PUFs suitable as hardware-bound secrets for key generation, device binding, and authentication in resource-constrained IoT devices.

### 3.2 Hash-based One-Way Accumulator

A one-way hash accumulator, proposed as an efficient alternative to digital signatures in authentication protocols, provides a compact, non-invertible representation of a set of elements. In such an accumulator, items are combined using a quasi-commutative operator, so the final accumulated value does not depend on the order of insertion.

Formally, a function

$$f : A \times B \rightarrow A$$

is said to be quasi-commutative if, for all  $a \in A$  and for all  $b, c \in B$ ,

$$f(f(a, b), c) = f(f(a, c), b). \quad (3.1)$$

Nyberg [13] introduced a fast accumulated hashing method suitable for lightweight systems. Let  $N$  denote the maximum number of accumulable items, and let  $h : \{0, 1\}^* \rightarrow \{0, 1\}^\ell$  be a cryptographically secure hash function. Given a set of items  $Y = \{y_1, y_2, \dots, y_m\}$ ,  $m \leq N$ , each item is first hashed as  $Y_i = h(y_i)$ ,  $i = 1, 2, \dots, m$ . The accumulated hash is then computed as:

$$T_{\text{Acc}} = H(s, Y) = s \bigodot_{i=1}^m h(y_i), \quad (3.2)$$

where  $s$  is an initialisation seed and  $\bigodot$  represents the bitwise AND operator. Since the AND operator is commutative and the hash function is one-way, the accumulator inherits

quasi-commutative and one-way properties.

**Membership Verification:** To verify whether an item  $y_i$  belongs to the accumulated set, compute  $Y_i = h(y_i)$ , and check whether  $Y_i \odot T_{\text{Acc}} = T_{\text{Acc}}$ .

The security of the accumulator depends on the difficulty of forging a value  $Y_i$  that passes verification. For a security parameter  $t$ , the required length  $r$  of the accumulated hash is given by  $r = N \times e \times t$ , where  $e$  is Euler’s (Napier’s) constant.

### 3.3 Core Concept of Anonymity Preserving

In IoT security, anonymity is defined by two key properties: **Untraceability** (concealing the sender’s real identity) and **Unlinkability** (preventing the correlation of multiple sessions to a single device). While simply encrypting a static ID protects the credential itself, it does not prevent traffic analysis; if the encrypted value remains constant, an adversary can track the device’s activity patterns.

To achieve robust anonymity, modern schemes utilise **Dynamic Pseudonyms**. Instead of a static identifier, the device generates a session-specific alias ( $PID$ ) using a shared secret or random nonce, e.g.,  $PID_{i+1} = H(ID \parallel Nonce_i)$ . To an external observer, each authentication request appears as a fresh, random string of bits, completely uncorrelated with previous sessions.

A critical challenge when using dynamic pseudonyms is **desynchronisation**. If the message carrying the new nonce is lost in transit, the device retains the old nonce while the server has already advanced to the new one. Their pseudonym computations then diverge, and all subsequent authentications fail. To handle this gracefully, a dual-state storage approach retains both the current and the immediately previous state at the server, allowing recovery without re-enrolment.



# Chapter 4

## DAuth: The Base Authentication Scheme

### 4.1 Overview of DAuth

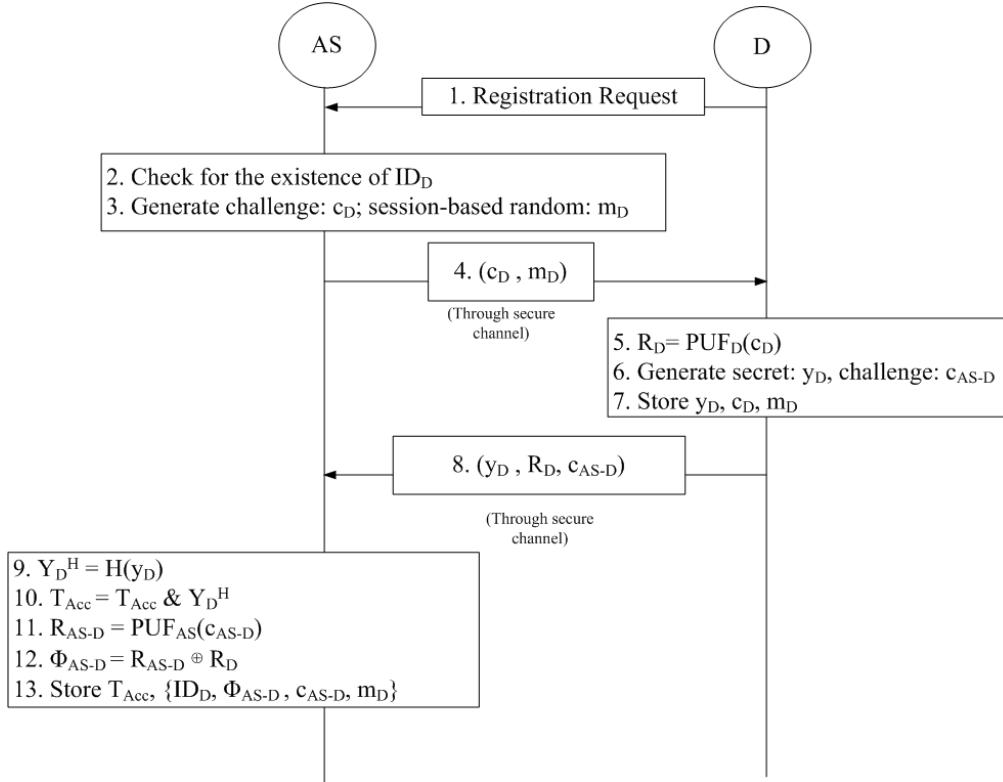
DAuth [1] is designed for multihop IoT networks where newly joined nodes must authenticate themselves securely before communicating with the gateway. Instead of relying on a centralised authority, the gateway distributes the authentication workload by selecting a set of already-authenticated IoT nodes to operate as Authentication Servers (AS). Let this set be represented as  $\mathcal{AS}' = \{AS_1, AS_2, \dots, AS_n\}$ , where the gateway shares a long-term secret key  $K_{GW-AS}$  with each  $AS_i$ .

When a new device  $D$  joins the network, the gateway (through the parent node) provides  $D$  with the address and identity  $ID_{AS}$  of the responsible authenticator. Importantly, the AS may lie at a single-hop or multi-hop distance, and does not need to be the parent node. This *decoupling* reduces the computational burden on both gateway and parent nodes, improving scalability in dense IoT environments.

The operation of the scheme is divided into two major phases: the **Node Enrollment Phase** and the **Node Authentication and Key Exchange Phase**.

#### 4.1.1 Node Enrollment Phase

In the enrollment phase, node  $D$  communicates with its AS over a secure channel. During enrollment,  $D$  generates its unique secret  $y_D$ , hashes it to obtain  $Y_D = H(y_D)$ , and registers this value with the AS. The AS incorporates  $Y_D$  into its hash-based accumulator using  $T_{Acc} = T_{Acc} \& Y_D$ . To avoid storing large CRP databases, the AS stores only a lightweight entry consisting of the challenge  $c_{AS-D}$  and a masked PUF response  $\Phi_D = R_{AS-D} \oplus R_D$ , while  $D$  stores its own challenge  $c_D$ . This significantly reduces memory usage and prevents direct leakage of raw PUF responses. Figure 4.1 illustrates this phase.



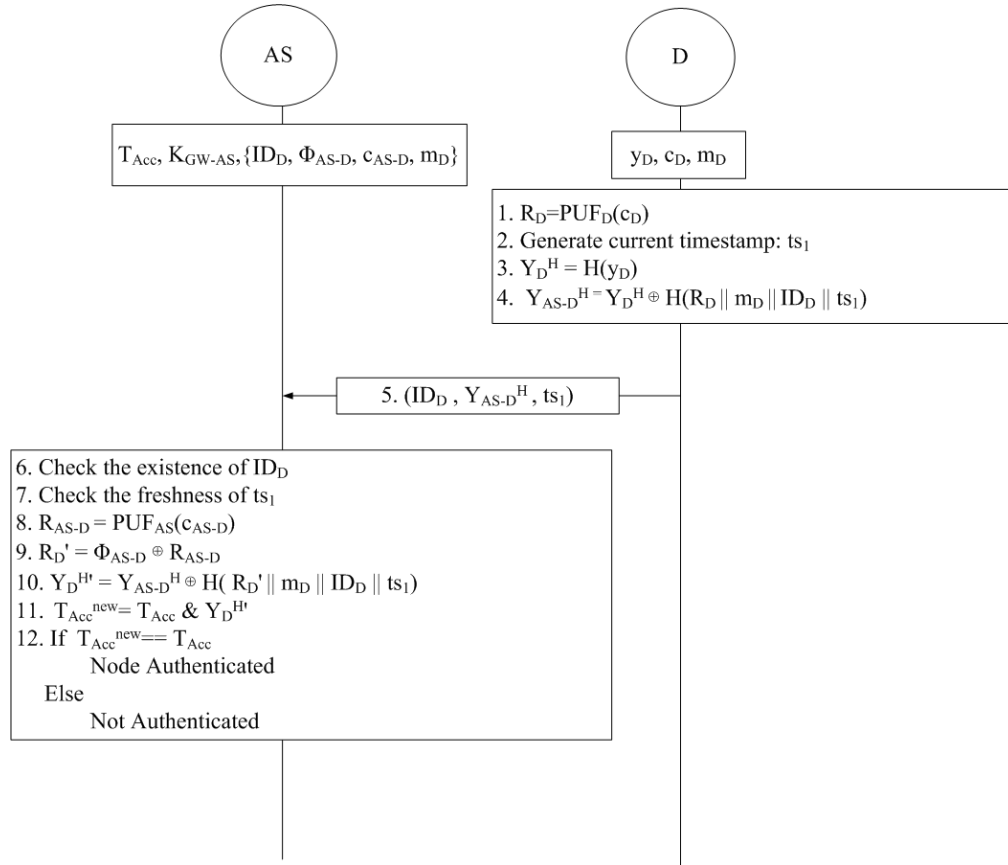
**Fig. 4.1:** Node Enrollment Phase of DAuth (adapted from [1]).

#### 4.1.2 Node Authentication and Key Exchange Phase

In the **Node Authentication Phase**, node  $D$  authenticates itself to the AS over an insecure public channel.  $D$  regenerates its PUF response  $R_D = PUF_D(c_D)$  and computes a mask  $H(R_D \| m_D \| ID_D \| ts_1)$ , where  $m_D$  is the current session-based random obtained during

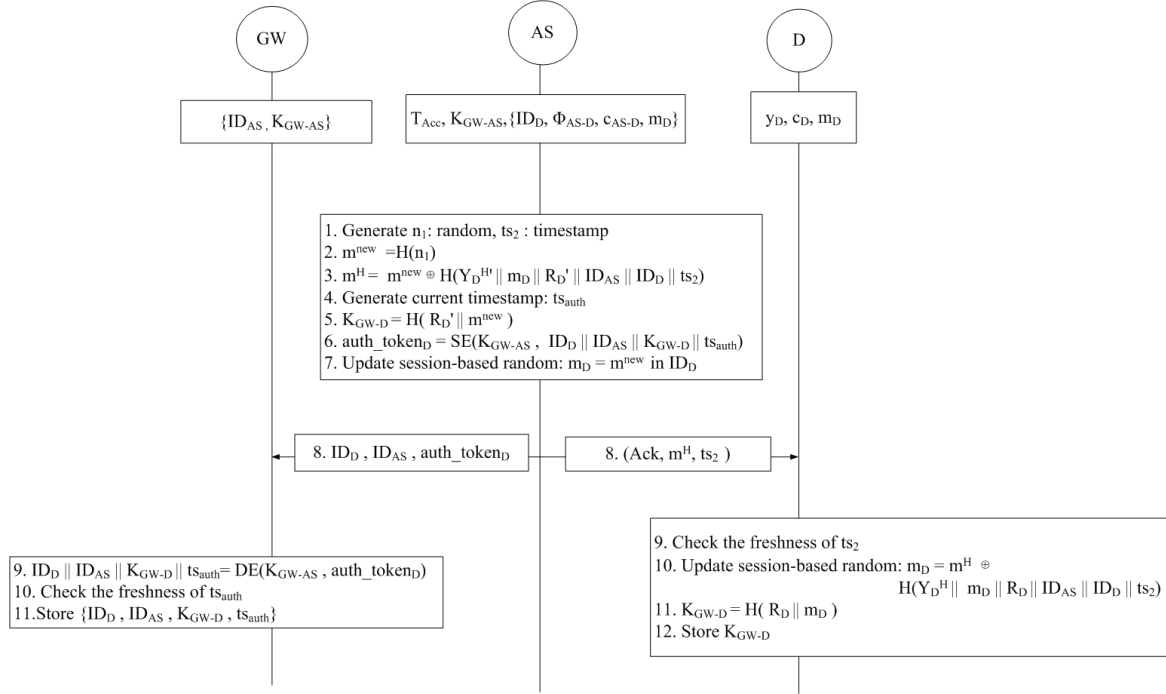
enrollment. The hashed witness  $Y_D^H = H(y_D)$  is XORed with this mask and transmitted to the AS along with the real device identity  $ID_D$  and timestamp  $ts_1$ .

Upon receiving the message, AS reconstructs  $R_D$  using  $\Phi_D \oplus R_{AS-D}$ , regenerates the mask, unmaskes  $Y_D^{H'}$ , and verifies authenticity by checking  $T_{Acc}$  &  $Y_D^{H'} = T_{Acc}$ . If equality holds, the AS confirms that  $D$  was previously registered. Figure 4.2 shows the authentication phase.



**Fig. 4.2:** Node Authentication Phase of DAuth (adapted from [1]).

The AS then generates a new session-based random  $m_{new} = H(n_1)$ , where  $n_1$  is freshly chosen, replacing  $m_D$  for the next session and providing forward secrecy. The session key for secure communication between the gateway and the node is computed as  $K_{GW-D} = H(R_D || m_{new})$ . AS constructs an authentication token containing  $ID_D$ ,  $ID_{AS}$ , timestamp  $ts_{auth}$ , and the session key  $K_{GW-D}$ , encrypted under  $K_{GW-AS}$ , and sends it to the gateway. Figure 4.3 illustrates this key exchange.



**Fig. 4.3:** Node Key Exchange Phase of DAuth (adapted from [1]).

#### 4.1.3 Privacy Gaps in DAuth

Although DAuth achieves decoupled and distributed authentication, PUF-based security, lightweight computation, low storage overhead, and dynamic key updates, it leaves two critical gaps unaddressed.

**Gap 1: Identity Exposure** The real device identity  $ID_D$  is transmitted in plaintext on the open channel in every authentication session. In the open wireless environments typical of IoT deployments, a passive attacker monitoring the channel can readily capture these identity transmissions. Even when all other packet fields carry opaque values, the presence of a plaintext  $ID_D$  is sufficient for an adversary to link multiple authentication attempts to the same device. Accumulated over time, such captures enable long-term device tracking, session linkage, topology inference, and targeted denial-of-service against specific nodes. In a multihop network, the exposure is amplified, since each intermediate hop introduces an additional vantage point from which the identity can be observed.

**Gap 2: Desynchronisation** DAuth stores only a single session nonce  $m_D$  at the AS.



If the key-exchange response carrying the new  $m_{\text{new}}$  to device  $D$  is lost,  $D$  retains the old  $m_D$  while the AS has already advanced to  $m_{\text{new}}$ . On the next authentication attempt,  $D$  constructs its authentication message using the stale  $m_D$ , but the AS uses  $m_{\text{new}}$  to verify; the masks no longer match, authentication fails, and the device must undergo full re-enrolment.

These two deficiencies motivate the proposed scheme, which hides device identities from the open channel, prevents session linkage, and adds desynchronisation resilience while preserving the lightweight and distributed character of DAuth.



# Chapter 5

## Proposed Authentication Scheme

### 5.1 System Model and Assumptions

The proposed scheme is designed to help nodes establish secure multihop communications with the gateway. To establish a secure communication, nodes should be authenticated by a trusted authentication server. A set of existing IoT nodes inside the network is entrusted with the responsibility of authentication by the gateway in a distributed manner. The gateway also shares a secret key with each of those responsible IoT nodes/authentication servers. The authentication server can be a parent or a non-parent node, which may be *decoupled*, preventing newly joined child nodes from overwhelming their parent with biased authentication load.

IoT nodes are assumed to be embedded with a PUF and are capable of performing various cryptographic operations, such as encryption/decryption, XOR, hashing, bitwise AND, and concatenation.

Nodes  $D$  and  $AS$  are assumed to be the newly joined node and its authentication server, respectively.  $GW$  represents the gateway symbolically.  $AS$  may be located at a single-hop or multi-hop distance from  $D$ . It is assumed that  $GW$  has already selected the authentication server set and has established a shared secret key with each member of the set. Let  $\mathcal{AS}' = \{AS_1, AS_2, \dots, AS_n\}$  be the set of authentication servers selected

by  $GW$ , such that  $AS \in \mathcal{AS}'$ .  $GW$  has a shared secret key  $K_{GW-AS}$  with  $AS$ . It is also assumed that  $D$ , whose distinct identifier is  $ID_D$ , possesses the address of its authentication server  $AS$  along with its unique identity  $ID_{AS}$ . Table 5.1 elaborates symbols and functions incorporated in the following scheme.

**Table 5.1:** Notation Summary.

| Symbol                  | Description                                                 |
|-------------------------|-------------------------------------------------------------|
| $D, AS$                 | Newly joined device and its authentication server           |
| $GW$                    | Gateway / root node                                         |
| $H(\cdot)$              | Collision-resistant hash function (SHA-256)                 |
| $SE(\cdot), SD(\cdot)$  | Symmetric encryption/decryption (AES-128)                   |
| $\parallel$             | Concatenation operation                                     |
| $\oplus$                | Bitwise XOR operation                                       |
| $\&$                    | Bitwise AND operation                                       |
| $c_D, c_{AS-D}$         | PUF challenges for $D$ (issued by $AS$ ) and $AS$ (by $D$ ) |
| $m_D$                   | Session-based random issued by $AS$ during enrolment        |
| $m_{curr}, m_{old}$     | Current and previous session-based randoms at $D$ and $AS$  |
| $y_D, R_D$              | Unique secret and PUF response of $D$                       |
| $Y_D^H$                 | Hashed unique secret: $H(y_D)$                              |
| $T_{Acc}$               | Accumulated authentication value at $AS$                    |
| $\Phi_{AS-D}$           | PUF-response binding: $PUF_{AS}(c_{AS-D}) \oplus R_D$       |
| $PID_{curr}, PID_{old}$ | Current and previous pseudonyms of $D$                      |
| $K_{GW-D}$              | Session key between $GW$ and $D$                            |
| $K_{GW-AS}$             | Long-term shared key between $GW$ and $AS$                  |
| $ts_1, ts_2, ts_{auth}$ | Timestamps for freshness verification                       |

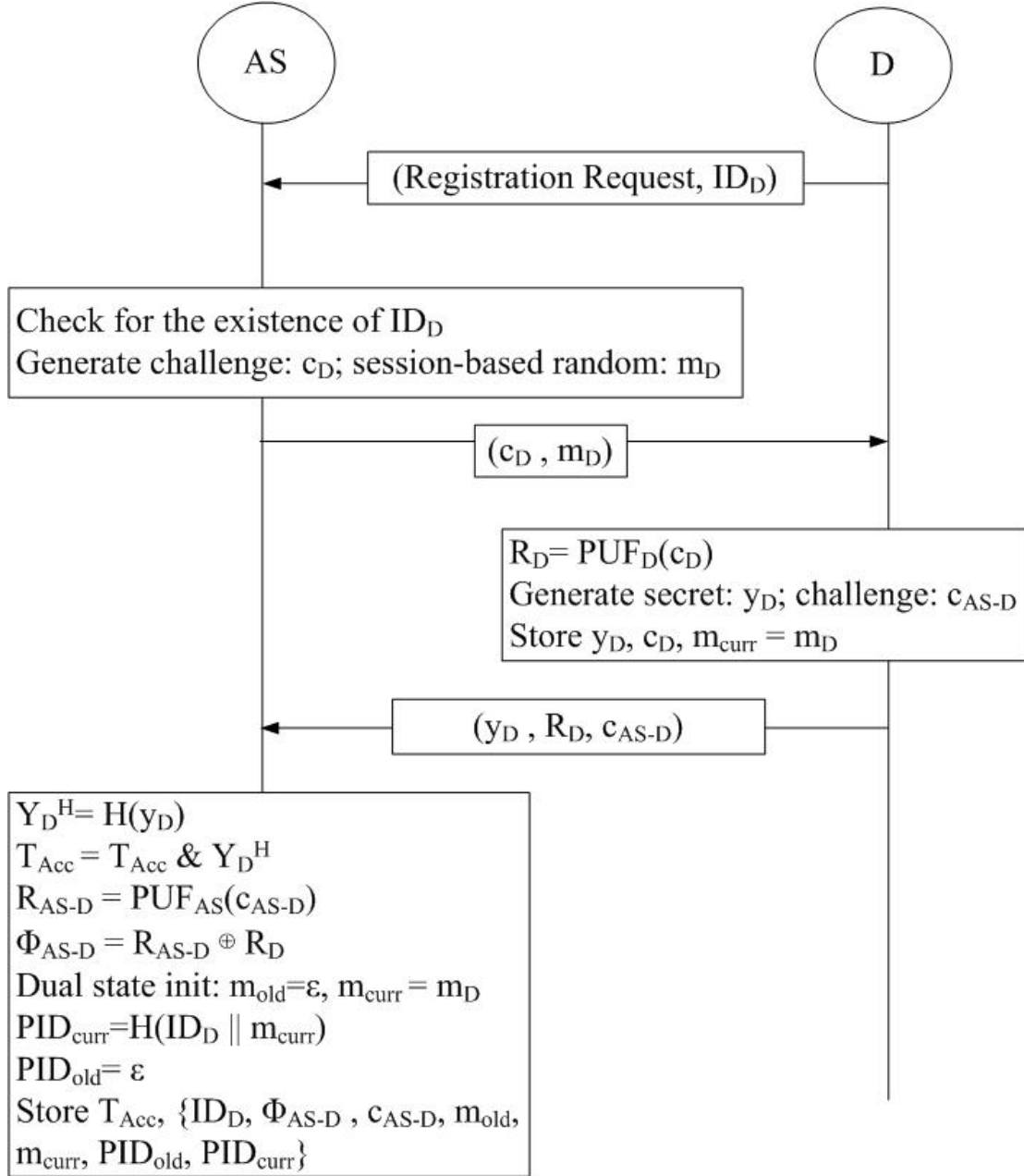
There are mainly three phases: node enrolment, node authentication, and node session key exchange. The following sections elaborate on these phases.

## 5.2 Node Enrolment Phase

The node enrolment phase is the first phase undergone by a newly joined node  $D$  with its authentication server  $AS$ . Before entering this phase, it is assumed that  $D$  has the required information regarding  $AS$ : the address and unique identity ( $ID_{AS}$ ) of  $AS$ .  $AS$  can lie at a single-hop or multi-hop distance from  $D$ . This phase is assumed to be executed in a secure environment by establishing an end-to-end encrypted channel between  $D$  and  $AS$ .

Every newly joined node executes this phase once until it disconnects from the network.

Figure 5.1 outlines the methodology of this phase.



**Fig. 5.1:** Proposed Scheme: Node Enrolment Phase with pseudonym initialisation.

$D$  initiates the phase by sending a registration request along with its unique identity  $ID_D$  to  $AS$  over the secure channel. As a response,  $AS$  generates a challenge  $c_D$  from the predefined challenge set  $\mathbb{C}_D$  and a session-based random  $m_D$ ; these are then forwarded

securely to  $D$ . With  $c_D$  as input to its embedded  $\text{PUF}_D$ ,  $D$  obtains a unique response  $R_D = \text{PUF}_D(c_D)$ . After this,  $D$  generates its own unique secret value  $y_D$ , also a random number; the security of the entire scheme critically depends on maintaining the confidentiality of  $y_D$ .  $D$  also picks a reverse challenge  $c_{AS-D}$  from the predefined challenge set  $\mathbb{C}_{AS-D}$  and sends  $(y_D, R_D, c_{AS-D})$  back to  $AS$  through the secure channel.  $D$  then stores  $\{y_D, c_D, m_D\}$ . Unlike [1], which stores  $m_D$  as a plain session random, the proposed scheme retains it as the initial *current* session random  $m_{\text{curr}}$ , seeding the pseudonym mechanism.

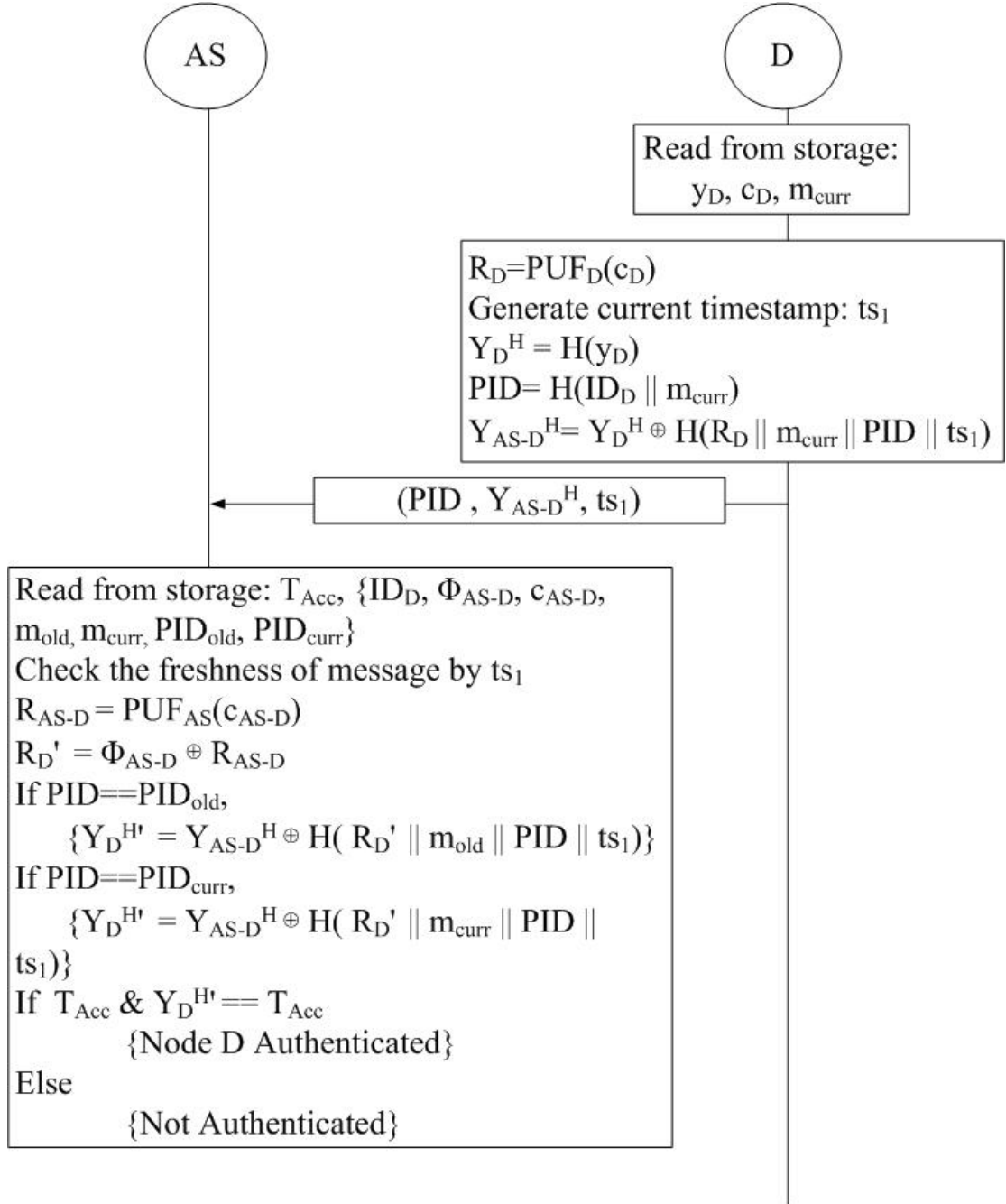
$AS$  inputs the secret  $y_D$  to the hash function  $H(\cdot)$ , generating tag  $Y_D^H = H(y_D)$ , and updates the accumulator as  $T_{\text{Acc}} = T_{\text{Acc}} \& Y_D^H$ . Further,  $AS$  generates  $R_{AS-D} = \text{PUF}_{AS}(c_{AS-D})$  and calculates  $\Phi_{AS-D} = R_{AS-D} \oplus R_D$ .

**Pseudonym Initialisation (new in proposed scheme):** The proposed scheme extends this base enrolment with pseudonym and dual-state initialisation to eliminate  $ID_D$  from all future open-channel messages. Since  $ID_D$  is transmitted only over this secure enrolment channel and must never appear on the open channel again,  $D$  will henceforth be identified by the rotating pseudonym  $PID = H(ID_D \| m_{\text{curr}})$  in all future authentication messages. Because  $H$  is one-way and  $m_{\text{curr}}$  is a private session random, successive pseudonyms are computationally unlinkable by a channel observer.  $AS$  pre-computes  $PID_{\text{curr}} = H(ID_D \| m_{\text{curr}})$  and sets  $PID_{\text{old}} = \perp$ ,  $m_{\text{old}} = \perp$ , forming the initial *dual state*.  $AS$  therefore stores  $\{T_{\text{Acc}}, ID_D, \Phi_{AS-D}, c_{AS-D}, m_{\text{curr}}, m_{\text{old}}, PID_{\text{curr}}, PID_{\text{old}}\}$ .

### 5.3 Node Authentication Phase

The node authentication phase is the second phase of the proposed scheme. Its main purpose is to authenticate  $D$ 's unique secret  $y_D$  with  $AS$ . Since this phase is executed over an open channel,  $y_D$  must be communicated in secure form, preventing it from being intercepted by adversaries. This security is established through PUF-based challenge-response, producing a secret key  $R_D$ , and the session-based random  $m_{\text{curr}}$ , incorporating unpredictability. Both  $R_D$  and  $m_{\text{curr}}$  are known only to  $D$  and  $AS$ . Figure 5.2 illustrates the

phase.



**Fig. 5.2:** Proposed Scheme: Node Authentication Phase with dual-state pseudonym lookup.

The phase is initiated by  $D$ .  $D$  regenerates  $R_D = \text{PUF}_D(c_D)$  from the stored challenge  $c_D$  obtained during enrolment.  $D$  then computes its current pseudonym  $PID = H(ID_D || m_{\text{curr}})$ , which is used in place of the real identity  $ID_D$  in the outgoing message.  $D$

produces the hashed secret  $Y_D^H = H(y_D)$  in XORed form as:

$$Y_{AS-D}^H = H(y_D) \oplus H(R_D \| m_{\text{curr}} \| PID \| ts_1). \quad (5.1)$$

The XOR mask is the hash of  $R_D$ , the session-based random  $m_{\text{curr}}$ , the pseudonym  $PID$ , and the current timestamp  $ts_1$ . The timestamp is used to provide replay protection to the XORed value  $Y_{AS-D}^H$ . The complete message  $(PID, Y_{AS-D}^H, ts_1)$  is then sent to  $AS$ .

After receiving the message,  $AS$  checks for the existence of an entry matching the received  $PID$ . In a departure from [1],  $AS$  performs a **dual-state lookup**: it checks whether the received  $PID$  matches  $PID_{\text{curr}}$  or  $PID_{\text{old}}$  in its records. A match on  $PID_{\text{old}}$  indicates that  $D$  missed the Key Exchange Phase delivery in the previous session and still holds the prior  $m_{\text{curr}}$ ; in this case,  $AS$  uses the stored  $m_{\text{old}}$  for the subsequent verification, recovering from the desynchronisation without requiring re-enrolment.  $AS$  verifies the freshness of  $ts_1$ , reconstructs  $R'_D$  as  $\Phi_{AS-D} \oplus \text{PUF}_{AS}(c_{AS-D})$ , regenerates the same XOR mask, unmaskes  $Y_D^{H'}$ , and then performs the bitwise AND check:

$$T_{\text{Acc}}^{\text{new}} == T_{\text{Acc}} \ \& \ Y_D^{H'} \stackrel{?}{=} T_{\text{Acc}}. \quad (5.2)$$

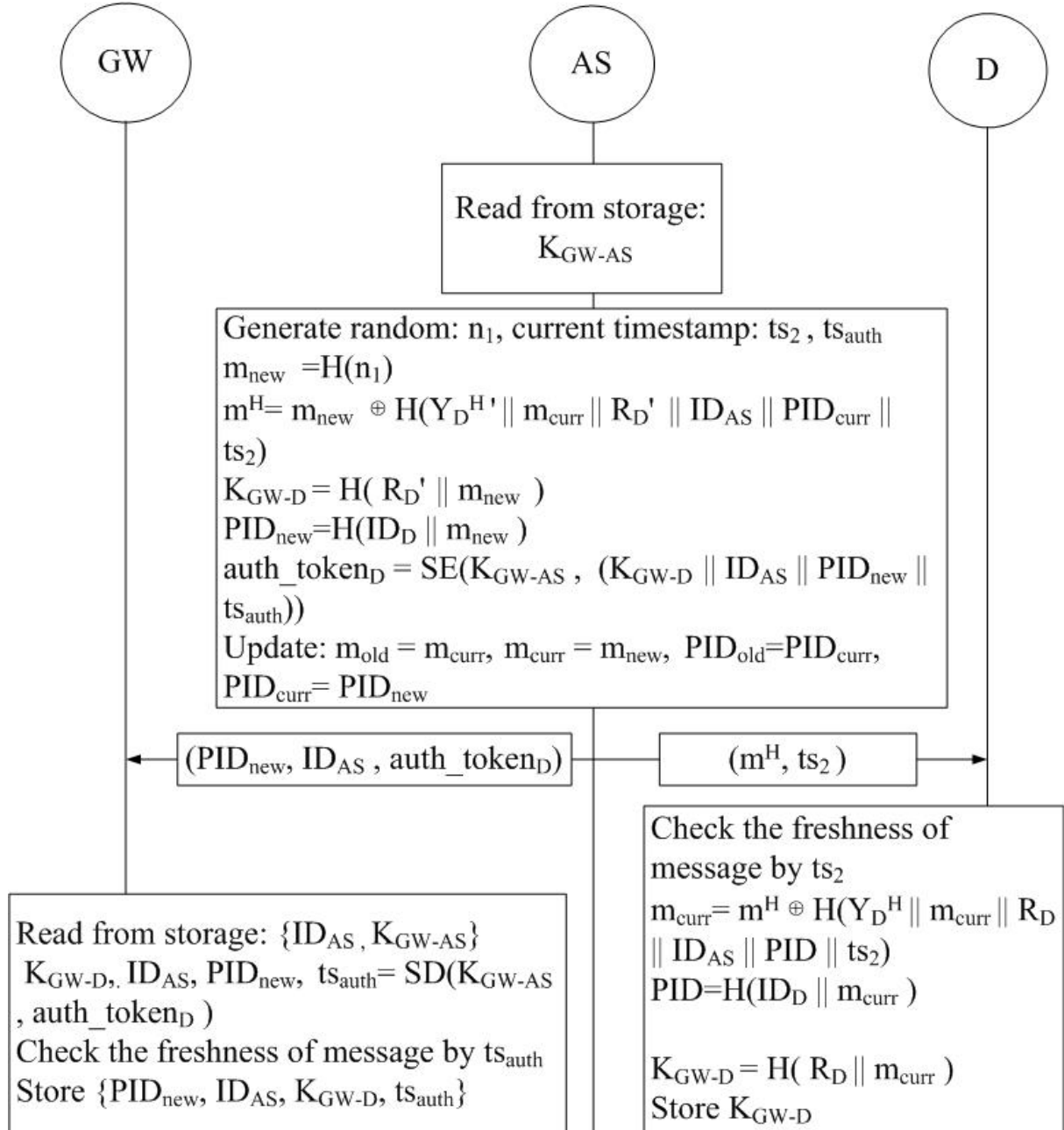
If  $T_{\text{Acc}}^{\text{new}} == T_{\text{Acc}}$ , it proves that the unique secret of  $D$  is already a legitimate member of  $AS$ . With the successful operation, the authentication of  $D$  is completed by  $AS$ , and  $AS$  proceeds to the Key Exchange Phase.

## 5.4 Node Session Key Exchange Phase

This phase is executed by  $AS$  to inform  $GW$  about the authentication of the newly joined node  $D$  at the end of the authentication phase.  $AS$  uses an `auth_tokenD` to deliver the session key  $K_{GW-D}$  to  $GW$ , facilitating session key generation between  $GW$  and  $D$ . The proposed scheme extends [1] by additionally deriving a fresh pseudonym  $PID_{\text{new}} = H(ID_D \| m_{\text{new}})$



at the end of each session and carrying it within  $\text{auth\_token}_D$ , so that  $GW$  can update its pseudonym mapping for  $D$  and successive sessions remain unlinkable. Figure 5.3 shows the phase.



**Fig. 5.3:** Proposed Scheme: Node Session Key Exchange Phase with pseudonym rotation.

$K_{GW-D}$  is calculated by  $AS$  using the hashed output of  $R'_D$  and  $m_{\text{new}}$ :

$$K_{GW-D} = H(R'_D \| m_{\text{new}}). \quad (5.3)$$

$m_{\text{new}}$  is computed by hashing a fresh random number  $n_1$  generated by  $AS$  as  $m_{\text{new}} = H(n_1)$ . Further,  $AS$  derives  $PID_{\text{new}} = H(ID_D \| m_{\text{new}})$ . The session key  $K_{GW-D}$ , the identities  $ID_{AS}$  and  $ID_D$ , the next pseudonym  $PID_{\text{new}}$ , and the authentication completion timestamp  $ts_{\text{auth}}$  are encrypted under the shared key  $K_{GW-AS}$  to form  $\text{auth\_token}_D$ .  $AS$  simultaneously sends  $(m_H, ts_2)$  to  $D$  and  $(PID_{\text{new}}, ID_{AS}, \text{auth\_token}_D)$  to  $GW$ , where the masked session seed is:

$$m_H = m_{\text{new}} \oplus H(Y_D^{H'} \| m_{\text{curr}} \| R'_D \| ID_{AS} \| PID_{\text{curr}} \| ts_2). \quad (5.4)$$

It then advances its records:  $m_{\text{curr}} \leftarrow m_{\text{new}}$ ,  $PID_{\text{curr}} \leftarrow PID_{\text{new}}$ , with  $m_{\text{old}}$  and  $PID_{\text{old}}$  retaining the superseded values.

Obtaining  $(m_H, ts_2)$ ,  $D$  first verifies the freshness of  $ts_2$ , then recovers  $m_{\text{new}}$  by re-generating the hash mask using its locally held  $Y_D^H$ ,  $R_D$ ,  $m_{\text{curr}}$ ,  $ID_{AS}$ ,  $PID_{\text{curr}}$ , and the received  $ts_2$ , and XORing with  $m_H$ .  $D$  establishes  $K_{GW-D} = H(R_D \| m_{\text{new}})$  and rotates its pseudonym to  $PID_{\text{curr}} = H(ID_D \| m_{\text{new}})$ , thus preparing itself for secure communication with  $GW$ .

$GW$  decrypts the token with  $K_{GW-AS}$ , verifies  $ts_{\text{auth}}$ , and extracts  $K_{GW-D}$  and  $PID_{\text{new}}$ . Unlike  $AS$ , the gateway maintains no dual state; it simply updates its single stored pseudonym for  $D$  to  $PID_{\text{new}}$  and associates the session key  $K_{GW-D}$  with it, so that  $D$ 's subsequent data frames carrying  $PID_{\text{new}}$  are recognised and served.

## 5.5 Desynchronisation Recovery

During the Key Exchange Phase,  $AS$  simultaneously dispatches two packets: one to  $GW$  carrying  $(PID_{\text{new}}, ID_{AS}, \text{auth\_token}_D)$ , and one to  $D$  carrying  $(m_H, ts_2)$ . Loss of either

packet leads to a different desynchronisation scenario, each handled without re-enrolment.

If the  $AS \rightarrow D$  packet  $(m_H, ts_2)$  is lost,  $D$  retains its stale  $m_{\text{curr}}$ , so the pseudonym it computes in the next session equals  $PID_{\text{old}}$  from  $AS$ 's perspective.  $AS$  recognises this via the dual-state lookup and resumes from the consistent prior state, restoring synchronisation without any re-enrolment overhead.

If instead the  $AS \rightarrow GW$  packet  $(PID_{\text{new}}, ID_{AS}, \text{auth\_token}_D)$  is lost,  $D$  has already advanced its local state: it holds  $m_{\text{new}}$  and  $PID_{\text{curr}} = PID_{\text{new}}$  after successfully receiving  $(m_H, ts_2)$ .  $GW$ , however, holds no valid session key for  $D$ . When  $D$  subsequently attempts data communication and  $GW$  cannot verify it,  $D$  re-initiates from the Authentication Phase using its current  $PID_{\text{curr}}$  (i.e.,  $PID_{\text{new}}$ ).  $AS$  matches this against  $PID_{\text{curr}}$  in its dual-state record and follows the normal authentication path, generating a fresh  $\text{auth\_token}_D$  and forwarding it to  $GW$ , which then correctly establishes the session key. No re-enrolment is required in this scenario either.

Algorithm 1 shows how the scheme adapts the Authentication and Key Exchange Phases to recover from the  $AS \rightarrow D$  loss scenario without re-enrolment.

The dual-state mechanism tolerates a single consecutive Key Exchange Phase loss. Two or more consecutive losses exhaust both slots and require re-enrolment, which is a deliberate trade-off between storage overhead and recovery depth.

---

**Algorithm 1** Desynchronisation Recovery
 

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**Precondition:**  $D$  holds stale  $m_{\text{curr}} = m_{\text{old}}$ ;  $AS$  holds  $(m_{\text{curr}}, m_{\text{old}}, PID_{\text{curr}}, PID_{\text{old}})$

*Authentication Phase*

- 1:  $PID \leftarrow H(ID_D \| m_{\text{curr}})$   $\triangleright$  stale  $m_{\text{curr}} \Rightarrow PID = PID_{\text{old}}$
- 2:  $D \rightarrow AS: (PID, Y_{AS-D}^H, ts_1)$
- 3: **if**  $PID = PID_{\text{curr}}$  **then**
- 4:      $AS$  verifies using  $m_{\text{curr}}$   $\triangleright$  normal path
- 5: **else if**  $PID = PID_{\text{old}}$  **then**
- 6:      $AS$  verifies using  $m_{\text{old}}$   $\triangleright$  desync detected
- 7: **else**
- 8:      $AS$  rejects  $D$
- 9: **end if**

*Key Exchange Phase (desync path)*

- 10:  $n_1 \leftarrow \text{rand}()$
- 11:  $m_{\text{new}} \leftarrow H(n_1)$
- 12:  $m_H \leftarrow m_{\text{new}} \oplus H(Y_D^{H'} \| m_{\text{old}} \| R'_D \| ID_{AS} \| PID_{\text{old}} \| ts_2)$
- 13:  $AS \rightarrow D: (m_H, ts_2)$
- 14:  $m_{\text{new}} \leftarrow m_H \oplus H(Y_D^H \| m_{\text{curr}} \| R_D \| ID_{AS} \| PID \| ts_2)$   $\triangleright D$  uses own  $m_{\text{curr}} = m_{\text{old}}$
- 15:  $m_{\text{old}} \leftarrow m_{\text{curr}}; m_{\text{curr}} \leftarrow m_{\text{new}}; PID \leftarrow H(ID_D \| m_{\text{curr}})$

**Result:**  $D$  and  $AS$  re-synchronised;  $PID$  rotated

---

# Chapter 6

## Security Analysis

### 6.1 Formal Security Analysis

The proposed scheme is translated into ProVerif [14] queries under the Dolev–Yao adversary model [15], where the adversary has full control of the public channel and can intercept, replay, modify, and inject messages. All 10 queries are satisfied. Figure 6.1 shows the verification output.

```
-----  
Verification summary:  
  
Query inj-event(DeviceEnrolled(id_D_2,id_AS_3)) ==> inj-event(DeviceEnrollmentStarts(id_D_2,id_AS_3)) is true.  
Query inj-event(DeviceAuthenticated(id_D_2,id_AS_3)) ==> inj-event(DeviceAuthenticationStarts(id_D_2,id_AS_3)) is true.  
Query inj-event(AuthenticationServerEnds(id_AS_3,id_D_2)) ==> inj-event(AuthenticationServerStarts(id_AS_3,id_D_2)) is true.  
Query inj-event(AuthenticationEndsFull(id_D_2,id_AS_3,R_D_2,m_D_2)) ==> inj-event(AuthenticationStartsFull(id_D_2,id_AS_3,R_D_2,m_D_2)) is true.  
Query event(DeviceKeyExDone(id_D_2,id_AS_3)) ==> event(DeviceKeyExStarts(id_D_2,id_AS_3)) is true.  
Query not attacker(SecretK_GW_D_Device[]) is true.  
Query not attacker(SecretK_GW_D_GW[]) is true.  
Query not attacker(SecretM_New[]) is true.  
Query not attacker(SecretID_D[]) is true.  
Weak secret SecretK_GW_D_Device is true.
```

**Fig. 6.1:** ProVerif verification output: all 10 queries satisfied.

### 6.1.1 Correspondence Queries (Q1–Q5)

Five correspondence queries verify that protocol steps occur in the correct order and that no adversary can skip or forge a step.

**Q1 (Enrolment correspondence):** A device can only be enrolled at the authentication server after it has explicitly initiated the enrolment process. This ensures a device can only be enrolled with its own participation.

**Q2 (Authentication, device side):** The device receives the server’s acknowledgement and the fresh timestamp  $ts_2$  only after it first sent a valid authentication request. This prevents replay of a stale acknowledgement.

**Q3 (Authentication, server side):** The authentication server completes its processing only after receiving a valid request from the device. This rules out an adversary forging a server-side completion event.

**Q4 (Two-round binding):** Q4 binds the Authentication and Key Exchange phases: the PUF response  $R_D$  and the session seed  $m_{\text{curr}}$  committed during authentication are proven to be the same values used in the key exchange. This rules out credential substitution between phases.

**Q5 (Key exchange correspondence):** The device recovers the new session seed  $m_{\text{new}}$  only after it legitimately sent the key exchange request.

### 6.1.2 Secrecy and Forward Secrecy (Q6–Q9)

**Q6–Q7 (Session key secrecy):** Two queries verify the secrecy of the session key  $K_{GW-D} = H(R_D || m_{\text{new}})$  from the device and gateway perspectives. Even though the adversary observes all traffic on the public channel, it cannot reconstruct  $K_{GW-D}$  because the PUF response  $R_D$  is never transmitted and  $m_{\text{new}}$  is always delivered XOR-masked.

**Q8 (Forward secrecy of  $m_{\text{new}}$ ):** This query confirms that the rotated session seed  $m_{\text{new}}$  remains secret. Even if an adversary recovers a current session key, it gains nothing about future sessions because each  $m_{\text{new}}$  is derived from an independently generated fresh

nonce.

**Q9 ( $ID_D$  anonymity):** After enrolment,  $ID_D$  is never sent over the open channel again. Every subsequent message carries the pseudonym  $PID = H(ID_D \| m_{\text{curr}})$ . ProVerif confirms that no attacker, even one who intercepts every message on the network, can recover  $ID_D$  from the observed traffic.

Anonymity is meaningful only if successive pseudonyms are unlinkable. Since Q8 proves  $m_{\text{new}}$  is secret, consecutive pseudonyms  $H(ID_D \| m_{\text{curr}})$  and  $H(ID_D \| m_{\text{new}})$  are computationally indistinguishable to any observer.

### 6.1.3 Offline Guessing Resistance (Q10)

**Q10 (Weak secrecy):** This weak-secrecy query confirms that the session key  $K_{GW-D}$  cannot be guessed from the public transcript even by an adversary with unlimited offline dictionary power. This protects against brute-force and dictionary attacks that try to recover the key from captured packets.

## 6.2 Informal Analysis

### 6.2.1 Replay Attack Resistance

The Enrolment Phase is executed over a secure channel and is immune to replay. The Authentication and Key Exchange Phases embed timestamps ( $ts_1$ ,  $ts_2$ ,  $ts_{\text{auth}}$ ) and the session-based random  $m_{\text{curr}}$  into the hash masks. A replayed message with a stale timestamp or stale  $m_{\text{curr}}$  will fail the freshness check at  $AS$ ,  $D$ , or  $GW$  respectively.

### 6.2.2 MITM Attack Resistance

The Enrolment Phase is conducted over an encrypted channel, which prevents MITM attacks. In the Authentication and Key Exchange Phases, the PUF response  $R_D$  (device-specific, unreproducible) and session random  $m_{\text{curr}}$  (private to  $D$  and  $AS$ ) are embedded

in every XOR mask. Without  $R_D$  and the current  $m_{\text{curr}}$ , no adversary can forge a valid masked tag  $Y_{AS-D}^H$  or unmask  $m_H$ .

### 6.2.3 Device Anonymity

$ID_D$  is transmitted only once, over the secure enrolment channel, and is never exposed on the open network again. All open-channel messages use  $PID = H(ID_D || m_{\text{curr}})$ , which is refreshed to  $H(ID_D || m_{\text{new}})$  after every key exchange. Because  $m_{\text{new}}$  is derived from a freshly generated nonce  $n_1$ , successive pseudonyms are computationally unlinkable.

### 6.2.4 Desynchronisation Recovery

Packet loss can occur at three distinct points in the Authentication and Key Exchange Phases, each handled without requiring re-enrolment.

If the acknowledgement and fresh timestamp  $ts_2$  sent by  $AS$  to  $D$  at the end of the authentication phase are lost in transit,  $D$  does not advance its internal state. Since  $AS$  has not yet performed any state update at this stage, both entities remain consistent. Upon timeout,  $D$  retransmits the authentication request with the same  $PID$  and  $ts_1$ , and the exchange proceeds normally.

The primary desynchronisation scenario arises when the Key Exchange Phase response  $(m_H, ts_2)$  is lost. In this case,  $AS$  has already advanced its state to  $m_{\text{new}}$  and rotated the pseudonym to  $PID_{\text{new}}$ , while  $D$  retains the previous  $m_{\text{curr}}$ . On the next authentication attempt,  $D$  constructs its pseudonym from the stale  $m_{\text{curr}}$ , producing  $PID_{\text{old}}$ .  $AS$  recognises this via the dual-state lookup and resumes from the consistent prior state.

If the token forwarded from  $AS$  to  $GW$  is lost,  $D$  re-initiates from the Authentication Phase; the dual-state storage at  $AS$  ensures that this re-authentication succeeds and a fresh token is forwarded to  $GW$ .

### 6.2.5 Forward Secrecy

$K_{GW-D} = H(R_D || m_{\text{new}})$ . Compromise of a session key does not reveal previous session keys because  $m_{\text{new}}$  is independently generated from a fresh nonce and  $R_D$  is PUF-derived.



# Chapter 7

## Performance Analysis

### 7.1 Theoretical Performance Analysis

#### 7.1.1 Computational Cost

Computational cost directly determines how efficiently an authentication scheme can be deployed on resource-constrained hardware. To obtain platform-native unit costs consistent with the hardware evaluation (Section 7.3), the per-primitive execution times reported by Lee et al. [16] are adopted, measured on a Raspberry Pi 4B (quad-core 64-bit ARM Cortex-A72 @ 1.5 GHz) using the MIRACL Core cryptographic library. The reference unit costs are: SHA-256 hash  $T_{\text{hash}} = 0.009 \text{ ms}$ , AES-128 encryption/decryption  $T_{\text{aes}} = 0.004 \text{ ms}$ , PUF evaluation  $T_{\text{puf}} = 0.12 \text{ ms}$ , and fuzzy extractor  $T_{\text{fe}} = 2.353 \text{ ms}$ . Following the standard code-offset construction [17], the dominant cost of the fuzzy extractor is one elliptic-curve scalar multiplication, hence  $T_{\text{fe}} = T_{\text{ecc}}$ . The cost of random-number generation and bitwise operations (XOR, concatenation) is negligible relative to these primitives and is therefore omitted, consistent with prior analyses [7]. The proposed scheme relies exclusively on PUF, hash, and AES operations, and employs no public-key or fuzzy-extractor primitive.

## Enrolment Phase

During node enrolment,  $D$  and  $AS$  each evaluate the PUF once, and two hash evaluations are performed (the unique-secret hash  $H(y_D)$  and the pseudonym initialisation  $H(ID_D || m_{\text{curr}})$ ), giving  $2T_{\text{puf}} + 2T_{\text{hash}} = \mathbf{0.258}$  ms. As shown in Table 7.1, this is marginally above DAuth [1] (0.249 ms), the difference being exactly one additional hash evaluation for pseudonym initialisation, a security feature absent from DAuth. Zhou et al. [7] incurs the highest enrolment cost (2.491 ms), dominated by the fuzzy-extractor generation function ( $T_{\text{fe}} = T_{\text{ecc}} = 2.353$  ms), whereas LAAKA [6] (0.027 ms) uses only hash operations and no PUF.

**Table 7.1:** Computational Cost Comparison: Enrolment Phase.

| Scheme          | Operations                                          | PUF | Cost (ms)    |
|-----------------|-----------------------------------------------------|-----|--------------|
| LAAKA [6]       | $3T_{\text{hash}}$                                  | ×   | 0.027        |
| Zhou et al. [7] | $T_{\text{puf}} + 2T_{\text{hash}} + T_{\text{fe}}$ | ✓   | 2.491        |
| DAuth [1]       | $2T_{\text{puf}} + T_{\text{hash}}$                 | ✓   | 0.249        |
| <b>Proposed</b> | $2T_{\text{puf}} + 2T_{\text{hash}}$                | ✓   | <b>0.258</b> |

$$T_{\text{hash}}=0.009 \text{ ms}, T_{\text{aes}}=0.004 \text{ ms}, T_{\text{puf}}=0.12 \text{ ms}, T_{\text{fe}}=T_{\text{ecc}}=2.353 \text{ ms}.$$

## Authentication and Key Exchange Phase

In the authentication and key exchange phases,  $D$  and  $AS$  each evaluate the PUF once,  $D$  performs six hash evaluations,  $AS$  performs five hash evaluations and one AES encryption, and  $GW$  performs one AES decryption. This yields  $2T_{\text{puf}} + 11T_{\text{hash}} + 2T_{\text{aes}} = \mathbf{0.347}$  ms, as shown in Table 7.2. The proposed scheme is second only to DAuth [1] (0.320 ms), the 0.027 ms overhead being attributable to three additional hash evaluations for pseudonym computation, dual-state lookup, and pseudonym rotation. LAAKA [6] (0.144 ms) achieves lower cost by performing only hash operations without any PUF evaluation; excluding the two PUF evaluations, however, the proposed scheme's remaining hash and AES operations total only  $11T_{\text{hash}} + 2T_{\text{aes}} = 0.107$  ms, below LAAKA's 0.144 ms, so the gap is attributable entirely to hardware-rooted PUF authentication rather than to protocol inefficiency. Zhou

et al. (2.608 ms) is the heaviest, owing to the fuzzy-extractor reproduction function ( $T_{fe} = 2.353$  ms).

**Table 7.2:** Computational Cost Comparison: Authentication & Key Exchange Phase.

| Scheme          | Operations                         | PUF          | Cost (ms)    |
|-----------------|------------------------------------|--------------|--------------|
| LAACA [6]       | $16T_{hash}$                       | $\times$     | 0.144        |
| Zhou et al. [7] | $T_{puf} + 15T_{hash} + T_{fe}$    | $\checkmark$ | 2.608        |
| DAuth [1]       | $2T_{puf} + 8T_{hash} + 2T_{aes}$  | $\checkmark$ | 0.320        |
| <b>Proposed</b> | $2T_{puf} + 11T_{hash} + 2T_{aes}$ | $\checkmark$ | <b>0.347</b> |

$T_{hash}=0.009$  ms,  $T_{aes}=0.004$  ms,  $T_{puf}=0.12$  ms,  $T_{fe}=T_{ecc}=2.353$  ms.

Combining both phases, the proposed scheme completes the full cycle in 0.605 ms, second only to DAuth [1] (0.569 ms) among PUF-based schemes. Both PUF-based schemes rely on four PUF evaluations ( $4T_{puf} = 0.480$  ms), which dominate their total costs. LAACA [6] (0.171 ms) achieves lower cost by forgoing PUF-based device authentication, while Zhou et al. [7] (5.099 ms) is the heaviest, with two ECC-based fuzzy-extractor operations ( $2T_{fe} = 4.706$  ms) accounting for over 92 % of its total cost. The proposed scheme thus incurs only a 0.036 ms premium over DAuth, corresponding to four additional hash evaluations across both phases, the price paid for anonymity and desynchronisation resilience.

### 7.1.2 Communication Cost

Communication cost is a core feature for analysing the performance of an authentication scheme, based on the number of message exchanges and their sizes (in bits). It is assumed that identities and timestamps are 32 bits, along with nonces, hashes, challenges, responses, and ciphertexts as 256 bits, ensuring a fair comparison.

### Enrolment Phase

During the node enrolment phase,  $D$  enrolls itself with  $AS$ , executing three message exchanges: (1)  $ID_D$  (32 b) from  $D$  to  $AS$ ; (2)  $c_D$  (256 b) and  $m_D$  (256 b) from  $AS$  to  $D$ ; (3)  $y_D$  (256 b),  $R_D$  (256 b), and  $c_{AS-D}$  (256 b) from  $D$  to  $AS$ . Total enrolment communication cost:

**1312 bits.** Table 7.3 shows the comparison.

**Table 7.3:** Communication Cost Comparison: Enrolment Phase.

| Scheme          | # Messages | Cost (bits) |
|-----------------|------------|-------------|
| LAACA [6]       | 3          | 2592        |
| Zhou et al. [7] | 5          | 1344        |
| DAuth [1]       | 3          | 1312        |
| <b>Proposed</b> | 3          | <b>1312</b> |

IDs/timestamps = 32 b; hashes/nonces/challenges/ciphertexts = 256 b.

### Authentication and Key Exchange Phase

The authentication and key exchange phase is executed using three message communications. Message 1 ( $D \rightarrow AS$ ):  $PID$  (256 b) +  $Y_{AS-D}^H$  (256 b) +  $ts_1$  (32 b). Message 2 ( $AS \rightarrow D$ ):  $m_H$  (256 b) +  $ts_2$  (32 b). Message 3 ( $AS \rightarrow GW$ ):  $PID_{\text{new}}$  (256 b) +  $ID_{AS}$  (32 b) +  $\text{auth.token}_D$  (256 b). Total: **1376 bits**. Table 7.4 shows the comparison.

**Table 7.4:** Communication Cost Comparison: Authentication & Key Exchange Phase.

| Scheme          | # Messages | Cost (bits) |
|-----------------|------------|-------------|
| LAACA [6]       | 3          | 2400        |
| Zhou et al. [7] | 4          | 2464        |
| DAuth [1]       | 3          | 928         |
| <b>Proposed</b> | 3          | <b>1376</b> |

IDs/timestamps = 32 b; hashes/nonces/challenges/ciphertexts = 256 b.

DAuth [1] requires 928 bits; the 448-bit increase in the proposed scheme directly reflects the widening of the device identifier from a 32-bit  $ID_D$  to a 256-bit pseudonym  $PID$ , which is the direct cost of adding device anonymity to all open-channel communications. The proposed scheme matches DAuth exactly in the enrolment phase (1312 bits), demonstrating that pseudonym initialisation adds no communication overhead over the DAuth at registration.

### 7.1.3 Storage Cost

Storage cost captures the persistent credentials each entity must retain to run authentication after enrolment, a decisive constraint for memory-limited IoT endpoints. To enable a consistent cross-scheme comparison, a single common rule is applied to every scheme: each entity is charged with the persistent state it must keep between sessions to authenticate again, excluding per-session ephemerals (random nonces, timestamps, session keys) and any value regenerated on demand (PUF responses, fuzzy-extractor secrets). All cryptographic values—hashes, nonces, challenges, PUF responses, ciphertexts, symmetric keys, and rotating pseudonymous identifiers—are normalised to 256 bits, while fixed real identities are taken as 32 bits, in line with the communication-cost assumptions. The per-entity footprints obtained under these assumptions are summarised in Table 7.5.

In LAAKA [6], the IoT device keeps its temporary identity  $TID_d$ , the fog temporary identity  $TID_f$ , the masked secret  $A_f$ , the shared anchor  $B_k$ , and its own authentication anchor  $A_d = h(ID_d || r_2)$ , which it must retain to recompute the authentication tag on every round (1280 bits). The fog server (FS) stores  $TID_d$ ,  $TID_f$ ,  $A_d$ , and  $B_k$  (1024 bits), while the registration authority (RA), as the root of trust, retains  $TID_d$ ,  $TID_f$ ,  $A_d$ ,  $A_f$ ,  $B_k$ , and the long-term master key  $K$  (1536 bits); it never stores the real identity  $ID_d$ , as the device registers by sending only  $A_d$ . In Zhou et al. [7], the gateway acts as the combined authenticator and router, maintaining per device the secret key, the identity, the dynamic identity, the sensor identity, the real sensor identity (used to refresh the sensor pseudonym), and a PUF challenge–response pair (1344 bits); the user device holds its identity, a masked credential, a helper string, the dynamic identity, and the sensor identity (1056 bits), while the sensor node stores only its own identifier (256 bits). In DAuth [1], the PUF-based device avoids long-term key storage yet still persists its unique secret  $y_D$ , the challenge  $c_D$ , and the current session nonce  $m_{\text{curr}}$  (768 bits); the PUF response  $R_D$  is regenerated on demand from  $c_D$ , but  $y_D$ —the secret enrolled with the accumulator—is stored. The AS retains  $T_{\text{Acc}}$ ,  $ID_D$ ,  $\Phi_{AS-D}$ ,  $c_{AS-D}$ , and  $m_{\text{curr}}$  (1056 bits), and the gateway keeps the session

key  $K_{GW-D}$  and  $ID_D$  (288 bits).

The proposed scheme inherits the same device footprint as DAuth (768 bits), because the rotating pseudonym  $PID = H(ID_D || m_{curr})$  is derived on the fly rather than stored. To support single-loss desynchronisation recovery, the AS additionally retains the previous nonce  $m_{old}$  and the dual pseudonym pair  $(PID_{curr}, PID_{old})$  on top of the DAuth state, raising its footprint to 1824 bits. Because the real identity is never exposed on the open channel, the gateway substitutes  $PID_{curr}$  for  $ID_D$  and stores it alongside  $K_{GW-D}$  (512 bits). As Table 7.5 shows, the proposed scheme and DAuth attain the lowest device-side storage (768 bits), roughly 40 % below LAAKA (1280 bits) and 27 % below Zhou et al. (1056 bits), making them well-suited for memory-constrained endpoints. The modest additional state introduced for anonymity and desynchronisation recovery is confined to the AS, a gateway-selected node with ample resources, and therefore imposes no burden on the constrained device.

**Table 7.5:** Per-device persistent storage cost (bits).

| Scheme          | Device     | AS / FS     | GW         | RA   | SN  |
|-----------------|------------|-------------|------------|------|-----|
| LAAKA [6]       | 1280       | 1024        | —          | 1536 | —   |
| Zhou et al. [7] | 1056       | —           | 1344       | —    | 256 |
| DAuth [1]       | 768        | 1056        | 288        | —    | —   |
| <b>Proposed</b> | <b>768</b> | <b>1824</b> | <b>512</b> | —    | —   |

## 7.2 Simulation-Based Performance Analysis

A simulation-based performance evaluation has been conducted using the COOJA simulator within the Contiki-NG framework. Three complementary studies are reported.

1. **Overall per-device comparison:** Proposed scheme vs. DAuth [1], LAAKA [6], and Zhou [7] in a fixed 100-node network (1 GW, 79 AS nodes with 2 active, 20 newly joined devices).
2. **AS pool-size study:** Active AS nodes varied from 2 to 10.

3. **Network scalability study:** Network size  $N$  varied across  $\{30, 50, 80, 100, 120\}$  nodes, with 20 % of nodes as newly joined devices.

All results are averaged over ten independent random seeds. Table 7.6 lists the common simulation parameters.

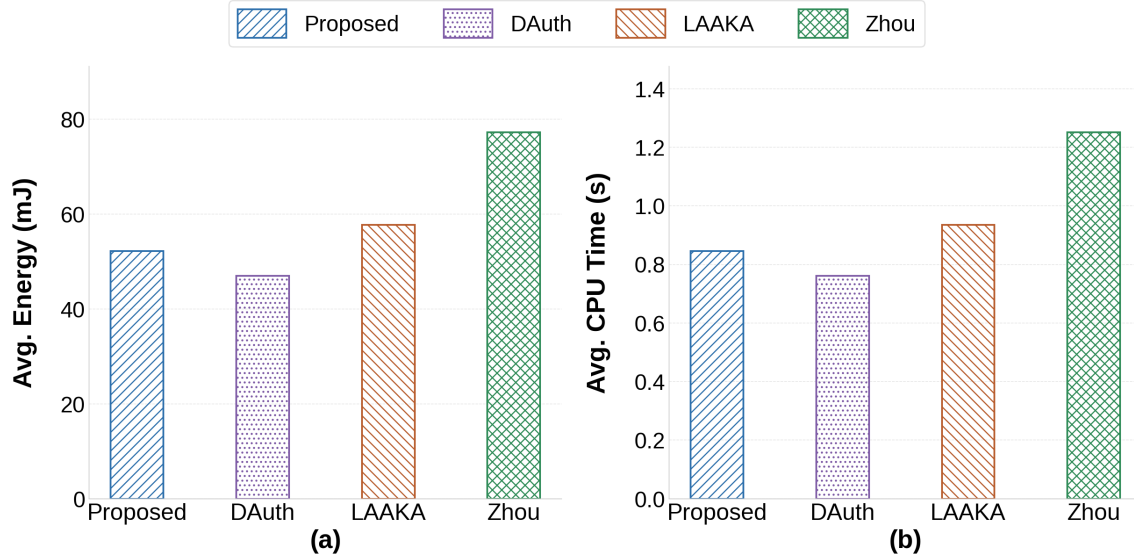
**Table 7.6:** Simulation Parameters.

| Parameter                | Value                                    |
|--------------------------|------------------------------------------|
| Operating System         | Contiki-NG                               |
| Mote Type                | Cooja Mote                               |
| Network Size             | 100 Nodes                                |
| Propagation Model        | UDGM (Distance Loss)                     |
| Application Layer        | CoAP                                     |
| Network Layer            | RPL Lite                                 |
| MAC Layer                | CSMA / IEEE 802.15.4                     |
| Simulation Time          | 75 Seconds                               |
| No. of Simulations       | 10 per configuration                     |
| Schemes compared         | Proposed, DAuth [1], LAAKA [6], Zhou [7] |
| Active AS (study ii)     | 2, 5, 10                                 |
| Network size (study iii) | $N \in \{30, 50, 80, 100, 120\}$         |

### 7.2.1 Overall Per-Device Cost

For 20 newly joined nodes in a fixed 100-node network with two active authentication servers, the proposed scheme demonstrated a consistent advantage over LAAKA and Zhou. As shown in Figure 7.1(a), the proposed scheme achieves a reduction of **9.5 %** and **32.4 %** in per-device energy consumption over LAAKA and Zhou et al., respectively. Similarly, a reduction of **9.5 %** and **32.4 %** is demonstrated in per-device CPU time (Figure 7.1(b)). DAuth [1] sits just below the proposed scheme in both metrics, as it requires no pseudonym operations; the proposed scheme pays a small overhead (approximately 8 %) for the pseudonym-rotation and dual-state mechanisms that deliver anonymity and desynchronisation resilience, both of which are absent from DAuth. Due to the incorporation of fewer message exchanges and lightweight hash-XOR operations, the proposed scheme minimises radio activity—the dominant energy cost in multihop IoT—compared to the

more message-intensive LAAKA and Zhou et al. schemes.



**Fig. 7.1:** Per-device mean total cost, all phases: (a) energy and (b) CPU time, 100-node network, 10 seeds.

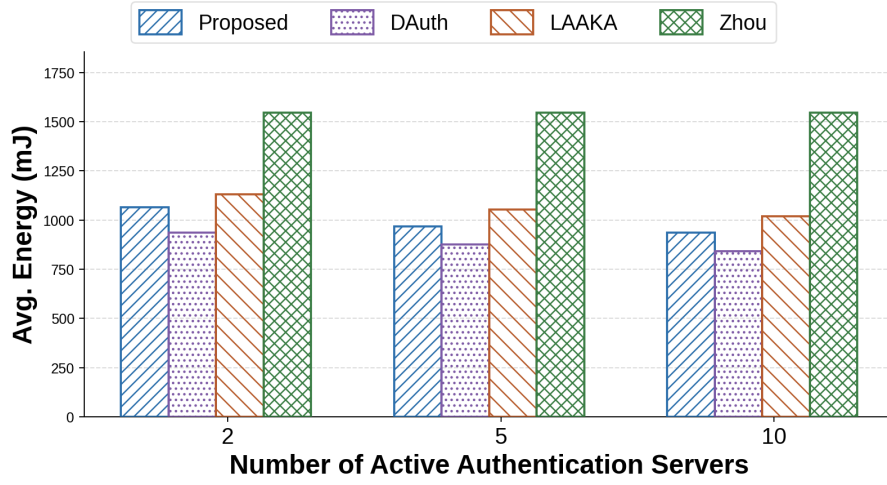
### 7.2.2 Authentication Server Pool Size

In the decoupled design, the gateway selects any capable node as the authentication server, so the size of the AS pool directly influences load distribution and overall network performance. The performance of the proposed scheme, DAuth, LAAKA, and Zhou is evaluated by varying the number of active AS nodes from 2 to 10.

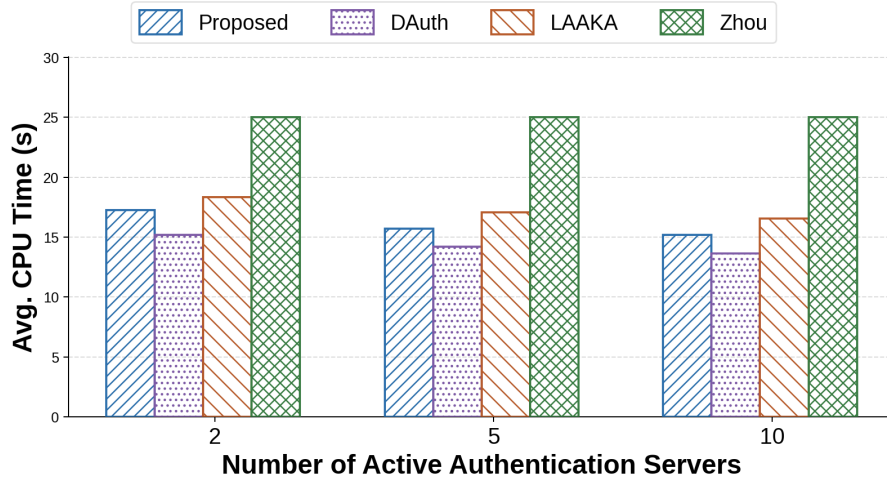
Considering the total energy consumption (Figure 7.2(a)), a reduction of approximately **6–8 %** is demonstrated by the proposed scheme over LAAKA at every AS count, and the same advantage holds for total CPU time (Figure 7.2(b)). Both distributed schemes also scale favourably with the pool, their total cost falling by roughly **11–12 %** as the pool grows from 2 to 10 servers, since additional authenticators absorb the join load of newly arrived devices. Zhou et al., included as a gateway-coupled baseline, has no authentication server pool and its cost remains essentially flat across all pool sizes, confirming that a centralised design cannot convert authentication server diversity into load relief. DAuth sits consistently just below the proposed scheme, reflecting the absence of pseudonym-



rotation overhead.



(a) Total energy vs. number of active AS nodes.



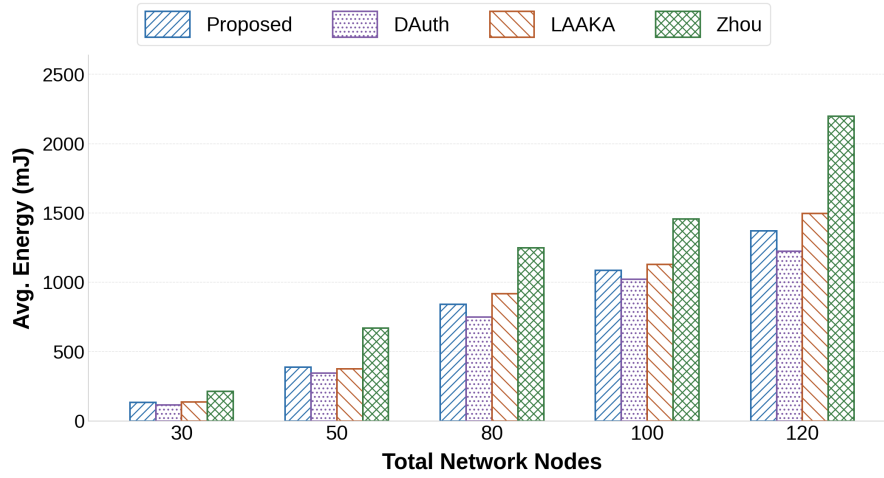
(b) Total CPU time vs. number of active AS nodes.

**Fig. 7.2:** Authentication cost vs. AS pool size (20 devices, 5 seeds).

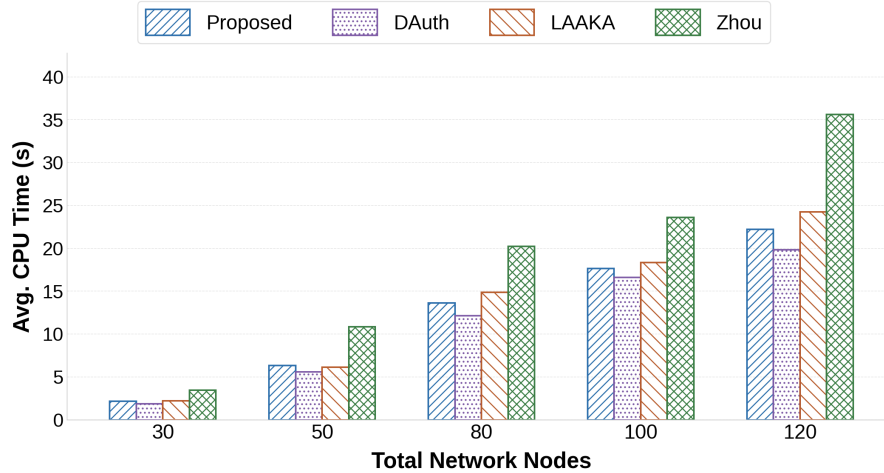
### 7.2.3 Network Scalability

The performance of authentication schemes is also analysed by considering their scalability with respect to total network size  $N$ , varied from 30 to 120 nodes, where 20% of nodes in each network are newly joined devices. Considering the total energy consumption (Figure 7.3(a)), at  $N = 120$  the proposed scheme demonstrates a reduction of **8.4 %** over LAAKA and approximately **44 %** over Zhou et al. Similarly, as shown in Figure 7.3(b),

a reduction of 8.4% and approximately 44% is demonstrated with respect to total CPU time over LAAKA and Zhou et al., respectively, at  $N = 120$ . LAAKA and Zhou et al. exhibit steeper energy and CPU growth with increasing  $N$  because their larger per-device message payloads and additional exchanges accumulate faster as the device count grows. DAuth and the proposed scheme both scale favourably, maintaining the lowest total cost across all network sizes; the proposed scheme incurs only a modest overhead above DAuth attributable to the additional pseudonym-rotation mechanism.



(a) Total energy vs. network size  $N$ .



(b) Total CPU time vs. network size  $N$ .

**Fig. 7.3:** Network scalability: total energy and CPU time vs.  $N$ .

#### 7.2.4 Desynchronisation Recovery

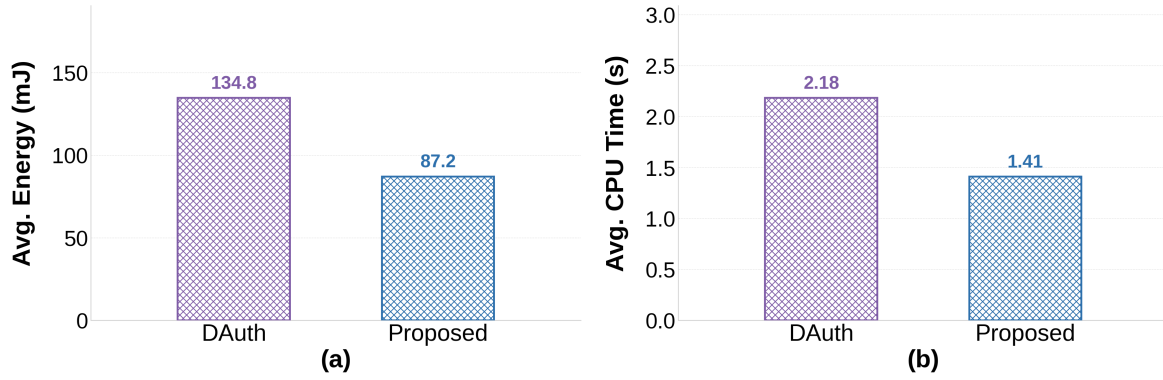
Desynchronisation affects protocols that rotate session state after each successful authentication: it arises when the packet delivering the updated state to one party is lost, leaving the sender on the new state while the receiver retains the old one. The next authentication attempt then fails from this mismatch, and the cost of recovery depends on the fallback mechanism the scheme provides.

In DAuth [1], desynchronisation is triggered when the key-delivery packet  $(m_H, ts_2)$  sent to the device at the end of the Key Exchange Phase is lost. The AS has already advanced its nonce to  $m_{\text{new}}$  while the device retains  $m_{\text{curr}}$ . As DAuth stores only a single nonce state, the device’s next attempt uses the stale  $m_{\text{curr}}$  and produces a pseudonym the server cannot recognise, so the request is rejected and recovery demands a complete re-enrolment, re-executing the PUF challenge–response exchange and reconstructing the hash-accumulator credential before authentication can resume. The proposed scheme shares the same protocol structure and is exposed to the identical trigger, but its AS retains a dual state,  $(m_{\text{curr}}, PID_{\text{curr}})$  and  $(m_{\text{old}}, PID_{\text{old}})$ . When the device re-authenticates with the stale pseudonym  $PID_{\text{old}}$ , the server recognises it via the dual-state lookup and completes a fresh authentication and key exchange without re-enrolment.

Both LAAKA [6] and Zhou et al. [7] also rotate their anonymous identifiers, but each retains the previous identifier rather than a single drifting secret. LAAKA derives  $TID_{d,\text{new}}$  from a fresh random each session while the fog server keeps the prior  $TID_d$ , and Zhou et al. have the gateway *add* the new  $DID_{i,\text{new}}$  to the user record while *retaining* the previous  $DID_i$ . In both schemes a device that misses the final message is therefore still recognised on its prior identifier and re-synchronises through an ordinary authentication round, without re-registration. As neither holds a single rotating secret that can drift irreconcilably between the parties, a like-for-like desynchronisation-recovery scenario does not apply to either, and both are excluded. The evaluation therefore contrasts DAuth and the proposed scheme only, which share the same trigger and threat model but differ in their

recovery mechanism.

Figure 7.4 reports the per-device cost of a recovery round, taken immediately after a desynchronisation event, over a 100-node RPL network (20 devices, 5 seeds). In DAuth, recovery requires a failed authentication, a full re-enrolment, and a fresh authentication and key exchange (134.8 mJ, 2.18 s). The proposed dual-state lookup instead recovers with a single re-authentication and key exchange (87.2 mJ, 1.41 s), cutting per-device recovery energy and CPU time by **35.3 %** over DAuth.



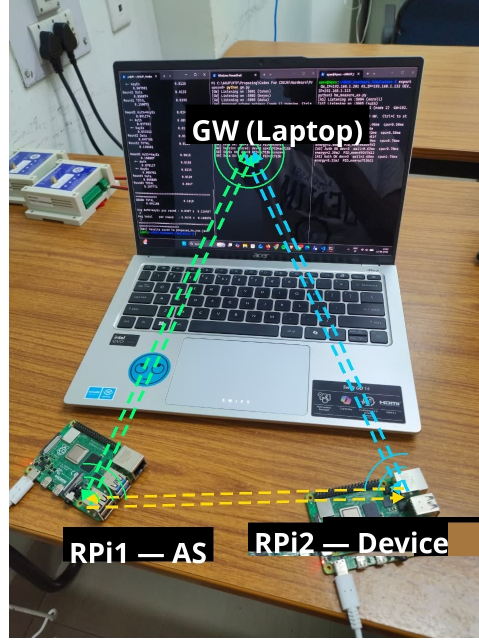
**Fig. 7.4:** Per-device desynchronisation recovery cost: (a) energy and (b) CPU time (DAuth vs. Proposed), over a 100-node RPL network (20 devices, 5 seeds).

### 7.3 Hardware-Based Performance Analysis

Hardware-based evaluation validates the feasibility of the proposed scheme by measuring its performance on physical devices under real operating conditions.

All four schemes (the proposed scheme, DAuth [1], LAAKA [6], and Zhou et al. [7]) are implemented on a **Raspberry Pi 4B** platform, equipped with a quad-core 64-bit ARM Cortex-A72 processor operating at 1.5 GHz, using the MIRACL Core cryptographic library for all hash, AES, and elliptic-curve operations. Identical hardware is used across all four schemes to ensure a fair comparison of energy consumption and CPU time. The implementation considers two Raspberry Pi 4B platforms as IoT nodes: one performing the role of the newly joined device, and the other functioning as the authentication server.

A Windows laptop is leveraged as the gateway. Each of the three entities maintains direct TCP communication with the others. Energy consumption is estimated using the active power profile of the Raspberry Pi 4B (3.8 W) multiplied by the measured wall-clock time per authentication round. The physical testbed is shown in Figure 7.5.

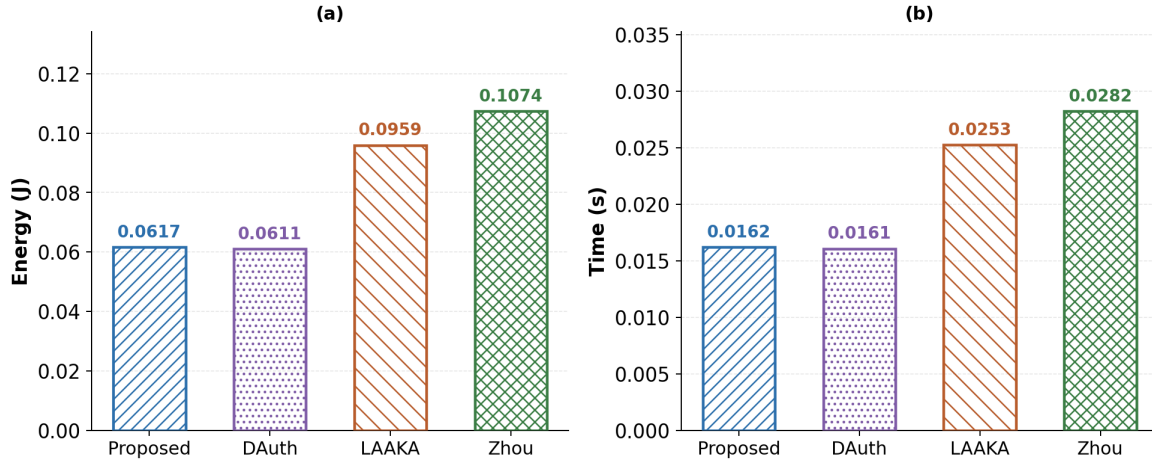


**Fig. 7.5:** Hardware testbed: two Raspberry Pi 4B boards (left: authentication server; right: device) connected to a Windows laptop (gateway).

As illustrated in Figure 7.6(a), the proposed scheme (**0.0617 J** per round) and DAuth [1] (**0.0611 J**) achieve virtually identical energy consumption, differing by less than **1 %**. This confirms that the additional SHA-256 evaluations for pseudonym rotation and the dual-state nonce lookup are sub-millisecond relative to the network round-trip time and introduce no measurable overhead over the non-anonymous baseline. Both outperform LAAKA [6] (0.0959 J) by approximately **36 %** and Zhou et al. [7] (0.1074 J) by approximately **43 %** in energy consumption. The higher cost of Zhou et al. arises from a nested gateway-to-sensor-node sub-exchange embedded within the authentication round that extends the effective device-side round-trip, whereas LAAKA’s two consecutive round-trips carry only lightweight hash-based payloads and therefore complete faster in absolute terms.

The CPU time results in Figure 7.6(b) follow the same ordering with identical percent-

age margins, since energy is directly proportional to wall-clock time under a fixed active power draw: the proposed scheme (0.0162s) and DAuth (0.0161s) perform equivalently, both remaining below LAAKA (0.0253s) and Zhou et al. (0.0282s) by approximately 36 % and 43 %, respectively. The ordering of schemes is consistent between the COOJA simulation and the hardware evaluation, confirming that the proposed scheme’s advantage reflects protocol design rather than platform-specific effects. The hardware evaluation thus validates that provable anonymity is achieved at no measurable cost penalty over the non-anonymous baseline on real IoT devices.



**Fig. 7.6:** Hardware-based performance: (a) energy consumption and (b) CPU time per authentication round on Raspberry Pi 4B (Proposed vs. DAuth vs. LAAKA vs. Zhou et al.).

# Chapter 8

## Conclusion

This report presented a lightweight anonymous authentication scheme for multihop IoT networks that extends DAuth [1] with two additions: per-session pseudonym rotation to hide  $ID_D$  from the open channel, and a dual-state mechanism to recover from desynchronisation without re-enrolment.

Formal verification in ProVerif confirms all 10 security queries under the Dolev–Yao model, covering authentication correspondence, session key secrecy, forward secrecy, anonymity, and offline guessing resistance.

COOJA/Contiki-NG simulations on 100-node networks show per-device energy reductions of **9.5 %** over LAAKA and **32.4 %** over Zhou et al. The dual-state mechanism is particularly significant in the desynchronised case: a single lost key-exchange packet forces DAuth into a full re-enrolment (134.8 mJ per recovery round), whereas the proposed scheme recovers through its dual-state lookup at 87.2 mJ, saving **35.3 %** in per-device recovery energy (and the same in CPU time) over DAuth.

Hardware experiments on Raspberry Pi 4B confirm that the proposed scheme attains virtually identical per-round cost to the non-anonymous DAuth baseline (within **1 %**), while reducing energy by approximately **36 %** over LAAKA and **43 %** over Zhou et al., demonstrating that the anonymity guarantee is achieved at no measurable overhead on real hardware.

Among the four compared schemes, the proposed scheme is the only one to simultaneously provide device anonymity, desynchronisation recovery, and decoupled distributed authentication at overhead suitable for resource-constrained IoT devices.

### 8.1 Future Work

Reducing the dual-state storage overhead at the authentication server while preserving desynchronisation resilience is a promising direction for further optimisation. The present

scheme also does not conceal the authentication server identity, as  $ID_{AS}$  appears in open-channel messages. Extending pseudonym rotation to authentication servers would provide full end-to-end identity unlinkability across the multihop network.



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